

Refresher Course

Assessment in Learning

- Part 1 - November 2022
- *Dr. Stephen Jay D. Co*



Topics for Discussion

- Basic Principles of Assessment
- Assessments of/as/for Learning
- Frames of Interpretations
- Sources of Evaluation
- Principles of High-Quality Assessments
- Instructional Objectives
- Domains of Learning
- Categories of Learning Outcomes

“The evaluation of pupils’ progress is a major aspect of the teacher’s job.”

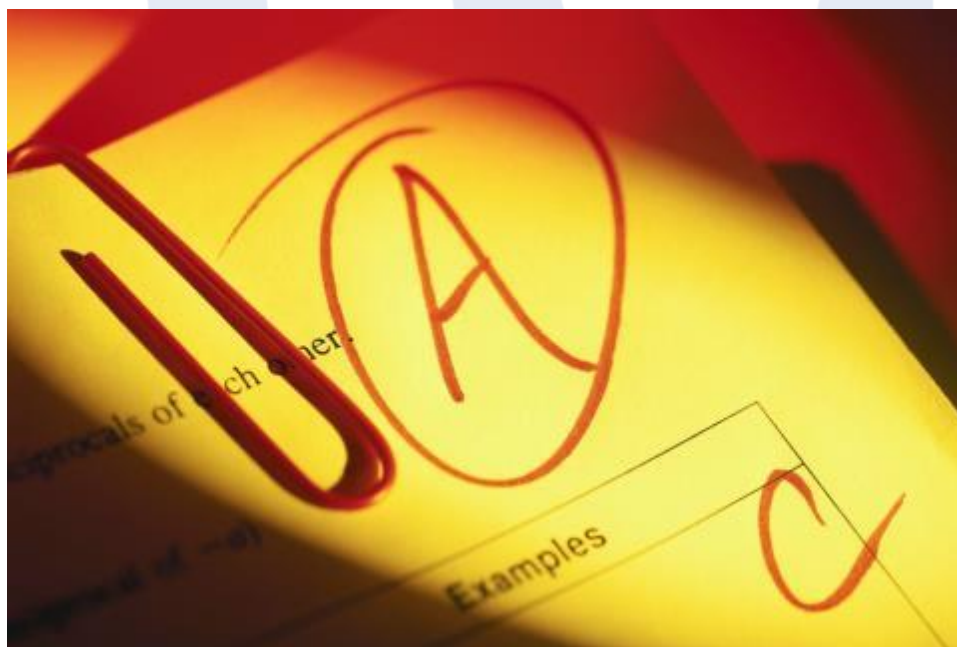


Evaluating Educational Outcomes
(Oriondo & Antonio)



Think Critically

- What are the differences among measurement, assessment and evaluation?





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What is Measurement?

Measurement

- The process of *determining a qualitative or quantitative attitude* of an individual or group of individuals that is of *academic relevance*.



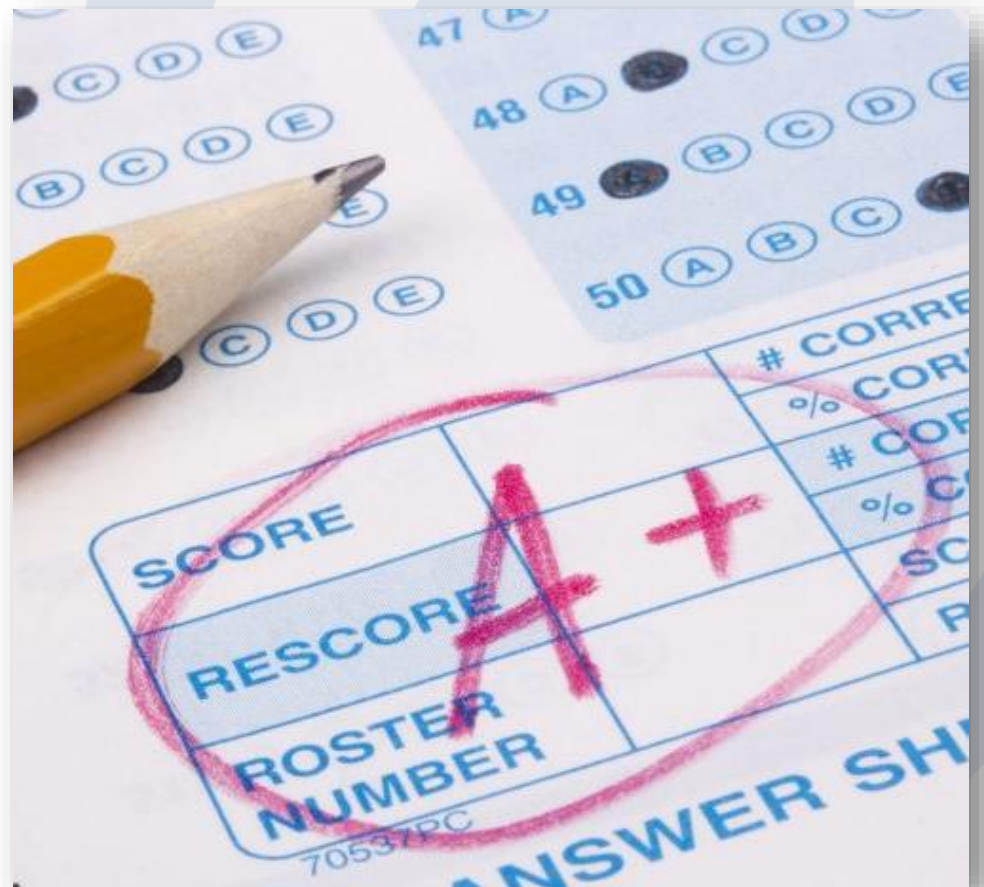


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What is Measurement?

Measurement

- The assignment of *numbers or other symbols* to certain characteristics of the objects of interest according to some specified rules in order to reflect quantities or properties





What is Measurement?

Measurement

- The quantification of what students learned through the use of tests, questionnaires, rating scales, checklists, and other devices.
- *‘How much does a student learn or know?’*

Name: _____ Date: _____

Kinds of Rocks Quiz

Sedimentary	Igneous	Metamorphic
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Directions: Write the word in the blank to tell what type of rock goes with each clue.

1. _____ This rock is formed from melted rocks.
2. _____ This rock is formed in layers.
3. _____ Fossils are often found in this rock.
4. _____ This rock is changed by heat and pressure.
5. _____ This rock was formed from pressure from the water in a lake or ocean pushing down on it.
6. _____ This rock may be formed from magma that hardens.
7. _____ This rock is often found near water.
8. _____ This rock may be formed from lava.
9. _____ Fossils are often found in this rock.
10. _____ You would most likely find this rock near a volcano.

Bonus Questions

What kind of rock?

1. Granite _____ 2. Marble _____



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What is Test?

Test

- Serves as a vehicle used to observe the student's attributes (i.e. written, oral or observation)



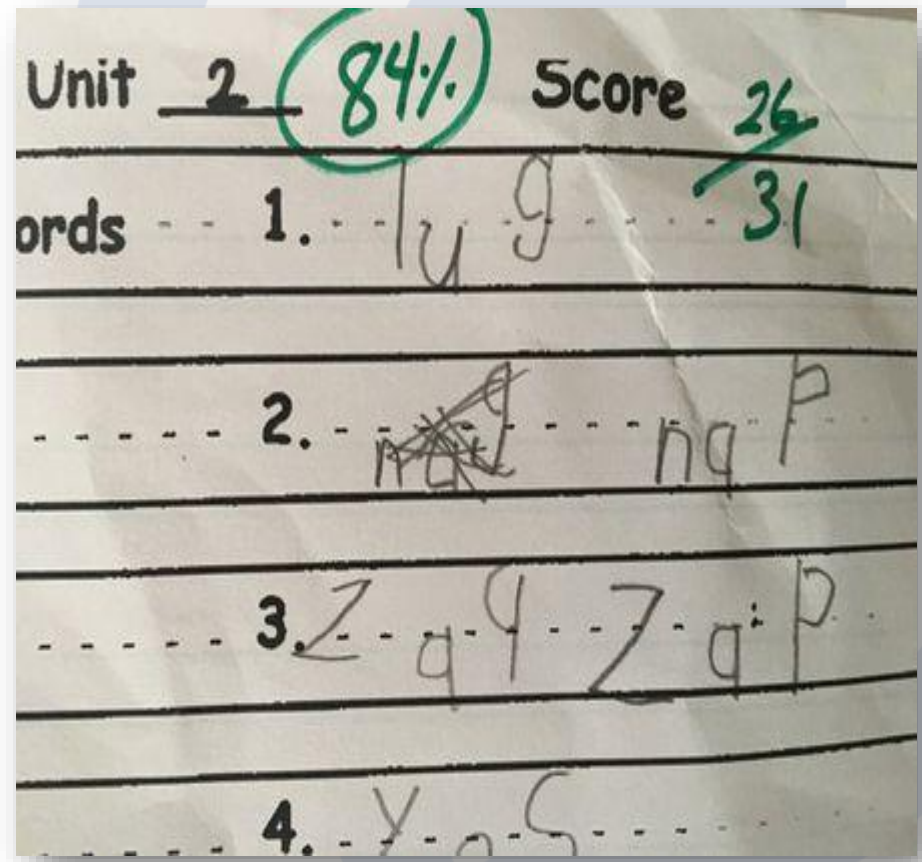


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What is Test Score?

Test Score

- Indication of what was observed through the test and it can be qualitative or quantitative in nature.



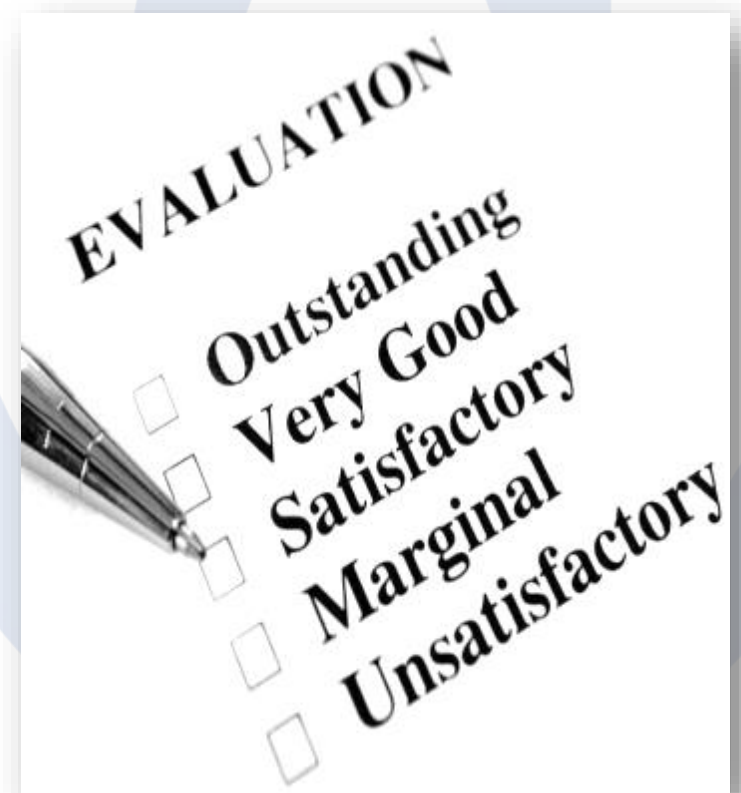


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What is Evaluation?

Evaluation

- It's the consideration of *evidence* in the light of *value standard* and in terms of the particular situations and the goals, which the group or individuals are striving to attain.



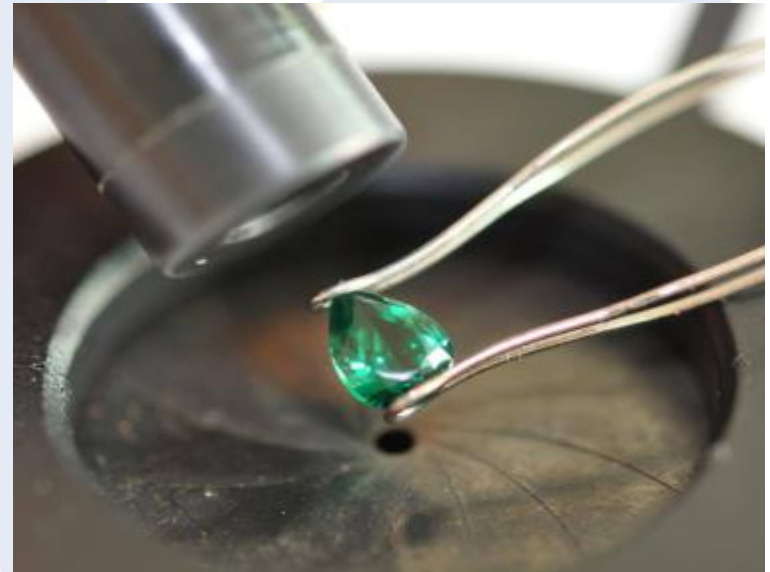
What is Evaluation?

Evaluation

- Is to make judgments about the quality or worth of something

Examples

- educational program
- work performance
- proficiency
- student attainments



What is Evaluation?

- Evaluation may answer such questions as...
 - How good is the level of achievement?
 - How good is the performance?
 - Have they learned enough?
 - Is their work good enough?



Assessment

The word 'assess'



- ▶ Latin verb 'assidere' meaning 'to sit with'
- ▶ In assessment, one should sit with the learner.
- ▶ It is something we do with and for students and not to students (Green, 1998)



Assessment

...is the
engine
which drives
student learning

(John Cowan)



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What is assessment?

- ▶ **Not the same as testing!**
- ▶ **An ongoing process to ensure that the course/class objectives and goals are met.**
- ▶ **A process, not a product.**
- ▶ **A test is a form of assessment. (Brown, 2004)**





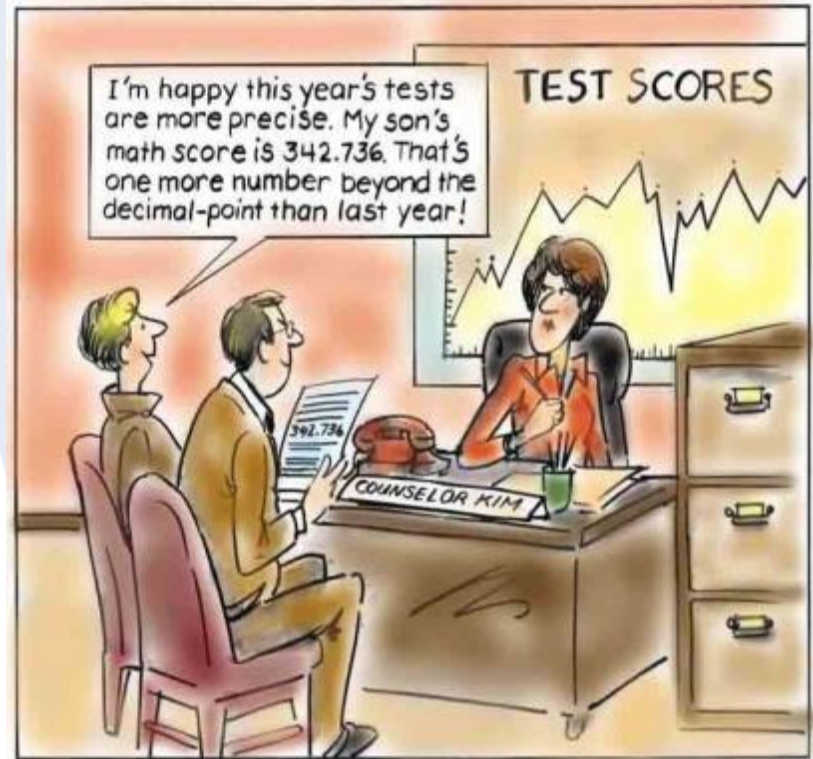
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What is Assessment?

Assessment

- Refers to a related *series of measures* used to *determine a complex attribute* of an individual or group of individuals.

Oosterhof (2001)





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What is Assessment?

Assessment

- Gathering and interpreting information about student level of attainment of learning goals.



Howard Gardner *on Assessment*

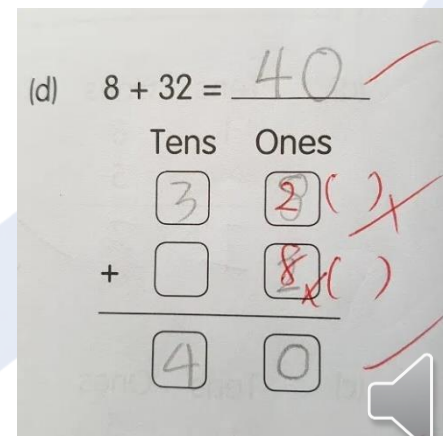
- American Developmental Psychologist born July 11, 1943.
- Professor of Cognition and Education at Harvard Graduate School of Education
- Best known for his *Theory of Multiple Intelligences*



***Do you agree with his ideas?
Write down a key idea that interest you.***

Assessments as Basis for Decision Making (Payne 2003)

1. The teacher reviews a work sample, showing some column additions are in error and there are frequent carrying errors.
2. He/She assigns simple problems on preceding pages, with consistent addition errors in some number combinations, as well as repeated errors in carrying from one column to another.



Assessments as Basis for Decision Making (Payne 2003)

3. He/She gives instruction through verbal explanation, demonstration, trial, and practice.
4. The student becomes successful in calculations made in each preparation step after direct instruction.
5. The student returns to the original page, completes it correctly, and is monitored closely when new processes are introduced.



Assessments as Basis for Decision Making (Payne 2003)

- Informed decision-making in education is very important owing to the obvious benefits it can bring about (Linn, 1999)
- Enhancement of student learning and development
- Feelings of competence
- Affective development - Self worth





Formative And Summative Assessments

- A. Formative Assessment
 - It is conducted to *monitor the instructional process* to determine whether learning is taking place as planned.
 - It is done frequently to determine who among the students have not attained the mastery of the learning task.



Formative And Summative Assessments

- A. Formative Assessment
 - activities that students and teachers undertake to get information about students' learning
 - giving students feedback which they can use to improve their performance



Uses of Formative Assessments for Students

1. Reinforcement

- Results of formative evaluation should inform the learners whether they have mastered the unit or not.



Uses of Formative Assessments for Students

2. Diagnosis

- The errors made by the students in the formative evaluation diagnose the weakness of the students as bases for remedial instruction.



Uses of Formative Assessments for Teachers

1. Handling Errors

- Teachers should handle errors made by the majority of the students. These errors should be reviewed by the class next session.





Uses of Formative Assessments for Teachers

2. Quality Control

- Records can be used as basis for remedial instruction and improvement of instruction by using different techniques and strategies, aids and devices.





- Results of summative test can be predicted on the basis of formative evaluation.





Formative and Summative Assessments

B. Summative Assessment

- is conducted at the *end of an instructional segment to determine if learning is sufficiently complete* to warrant moving the learner to the next segment of instruction.



Formative and Summative Assessments

B. Summative Assessment

- Judgement made on work that results in the student being awarded a grade, may include formative elements
- Assessment designed to be used to determine grades or marks



Summative Assessments

- Provides information on the product's efficacy (it's ability to do what it was designed to do)
- Typically QUANTITATIVE, using numeric scores or letter grades to assess learner achievement



Uses of Summative Assessments

1. To assign grades

- Results of summative test are rated, on the basis of a standard set for mastery.

2. To provide feedbacks to the students

- Grades will give the students on their performance in class especially if the test constructed is reliable



Uses of Summative Assessments

3. To compare outcomes of different groups

- The outcomes of the different teaching techniques and strategies, different teaching aids and devices and different types of students are compared.

Think Critically

- Formative Assessments should NOT be graded.



Caroline Wylie

Grading work meant for formative purposes can be “highly problematic,” especially if the goal is to get students to reveal what they don’t understand.



Kathryn Mitchell Pierce

“In this instance, the scores are for information purposes. They don’t ‘count’ in the final course or unit grade—they’re meant to be feedback on current progress,”



Scott Filkins

“If we say this is for practice and we hold you accountable for the practice, then it wasn’t really practice,” he said. “You almost want kids to make mistakes on formative assessments because that’s how you figure out your next teaching cues. Once we attach a grade, students try to hide their weaknesses.”



Principles of Assessment of/as/for Learning

- **Assessment FOR Learning (Formative)**

The primary purpose is to *give feedback* on teaching and student learning.

- a) It does not normally receive a grade, however low-weighted assessments are often used for feedback and feed-forward purposes (low-stakes assessment).
- b) It occurs both throughout and at the end of the learning process, although feedback is most successful for students when it is timely, valued and action-oriented, i.e., it feeds forward (Evans, 2013).



Principles of Assessment of/as/for Learning

- **Assessment FOR Learning (Formative)**

c) It is associated with terms such as dialogue, feedback, feed-forward and assessment criteria (Nicol, 2010).

d) Its role in decision-making is to help students (and/or teacher) to understand their own strengths and gaps in order to plan their future learning. It informs their actions for improvement of student learning or teaching quality.

e) It can have a powerful effect on student and teacher emotions; positive feedback can be engaging and motivating, negative feedback can be disengaging and demotivating.



Principles of Assessment of/as/for Learning

- **Assessment FOR Learning (Formative)**

f) Common examples include:

- written/oral feedback to students, teacher and students in a dialogue on the learning (Nicol, 2010),
- feed-forward advice (actions to improve next assessment),
- students giving feedback to other students, feedback to teacher on their teaching, and
- low-weighted in-class participation.



Principles of Assessment of/as/for Learning

- **Assessment OF Learning (Summative)**

The primary purpose is the demonstration of the achievement of student learning

- a) It is graded and is usually high stakes for students, teachers and schools (Earl & Katz, 2006).
- b) It usually occurs after a period of learning.
- c) It is associated with concepts such as measurement, competencies, standards, regulations and explicit criteria.



Principles of Assessment of/as/for Learning

- **Assessment OF Learning (Summative)**

- d) Its role in decision-making is to communicate students' performance to key stakeholders, often comparing them to other students (norm-referenced assessment) or to competencies (criterion-referenced assessment).
- e) It is often a key driver for student learning. It can be associated with student and/or teacher anxiety and workload but also can build student/ staff confidence when the outcome is to their satisfaction.
- f) Common examples are the quarterly/semester exam, essay, project, degree classifications.



Principles of Assessment of/as/for Learning

- **Assessment AS Learning (Formative)**

The primary purpose is to empower students to self-regulate and critically evaluate their learning and performance.

- a) It is ungraded assessment, with the rare exceptions of graded self-assessment.
- b) It occurs during the learning process and emphasizes the preparation of students for their future learning.
- c) It is associated with concepts such as self-monitoring, self-regulation, meta-cognition, learning and feedback (Evans, 2013; Lysaght & O'Leary, 2013; Sadler, 2010)



Principles of Assessment of/as/for Learning

- **Assessment AS Learning (Formative)**

d) Its role in decision-making is also to help students understand their own strengths and gaps in order to plan their future learning. It informs students' actions for improvement to their learning. It also supports the development of students' self-regulation skills.

e) It attempts to empower students, to give them confidence in their judgments and develop a sense of responsibility for their learning.



Principles of Assessment of/as/for Learning

- **Assessment AS Learning (Formative)**

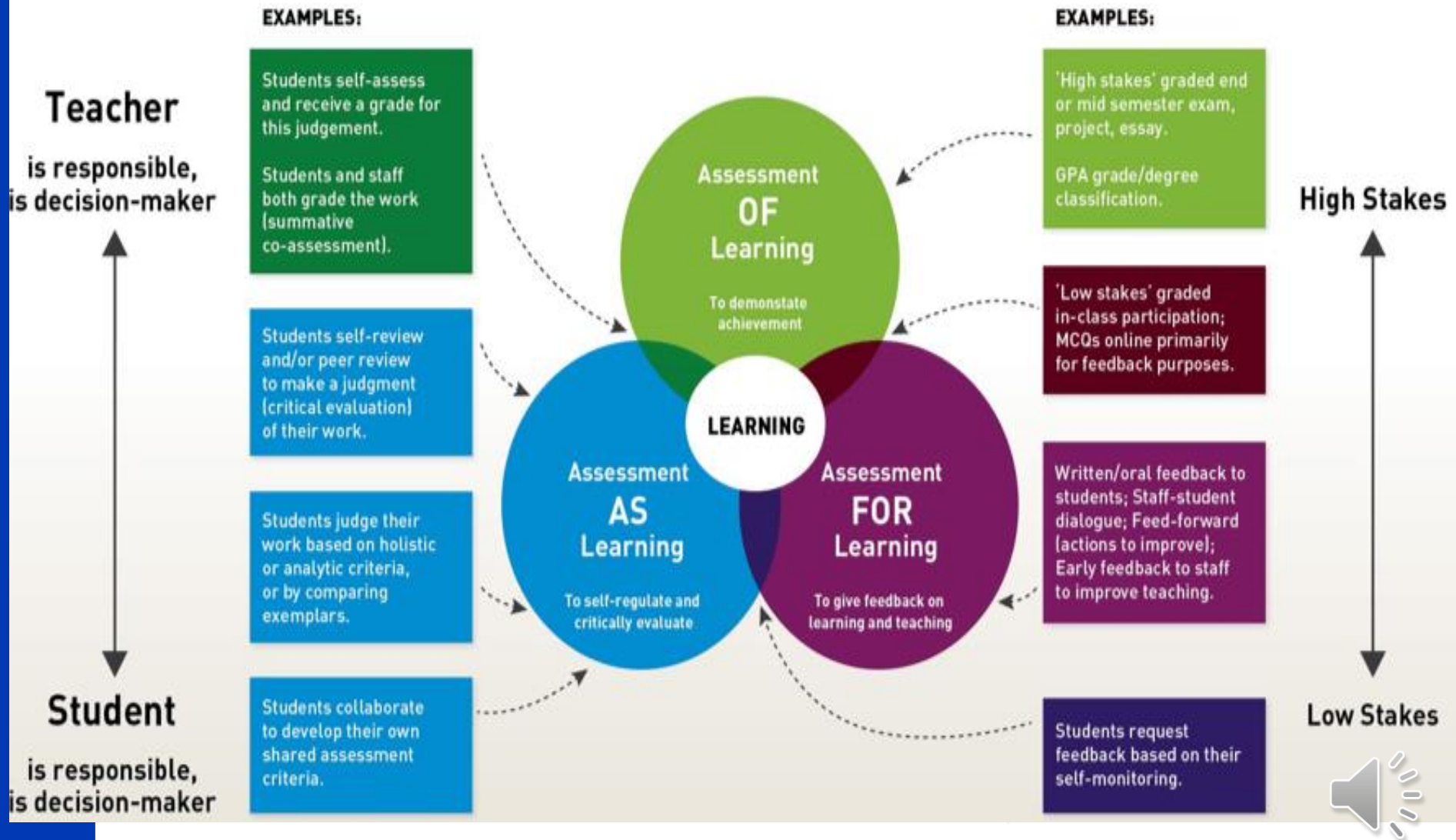
f) Common examples include

- students self- and peer reviewing their work against holistic or analytical criteria (Sadler, 2009),
- students using exemplars to judge their and other students' work (Carless et al., 2011), and
- students collaborating to develop their own shared assessment criteria (Evans, 2013).



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Principles of Assessment of/as/for Learning





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Measuring Achievement

- In principle, all important outcomes of education are measurable.
- To be important, an educational outcome must make a difference.
- If it makes a difference, the bases for measurement exist.

Can achievement be measured?



Suppose a student got a score of 25 in an examination, he or she may ask the questions:

“what does 25 mean?”

“Is this a high score or a low score?”



Frames of Reference for Interpreting Performance

1.) Ability-referenced

a student's performance is interpreted in light of that **student's maximum possible performance.**

interpretations are a comparison of a student's performance to that student's potential performance



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Ability-referenced

Examples of interpretation

“That’s about all this student can do”

“This student can do better if given more time”

The key of using ability as the frame of reference is having a good **estimate** of the student’s **maximum possible performance**.



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Ability-referenced

Standardized aptitude tests also provide teachers with estimates of students' ability, but these estimates are **general** and of **limited** use in interpreting a student's performance

The greatest limitation of ability estimates is that we usually do not know which abilities are prerequisite to the particular skills we are trying to teach.



Frames of Reference for Interpreting Performance

2.) Growth-referenced

performance is compared with the **student's prior performance.**

It a natural frame of reference in the classroom, since improvement in skills is highly relevant.



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Growth-referenced

Growth implies change over time, that is, change in performance between earlier and late measures of student performance.

Portfolio systems provide actual samples of earlier work to which present performance can be compared.



Frames of Reference for Interpreting Performance

3.) Norm-referenced

the interpretation is provided by comparing the **student's performance with the performance of others** or with the typical performance for the student.

How well a student did on a test is often described in terms of how others in the class did.



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Norm-referenced

Examples indicating

- A student can type faster than 60% of the students completing a typing class represents a norm-referenced interpretation of performance
- A student can do more push-ups 30% of other students the same age or
- A student can spell better than half the other students in the class.



Frames of Reference for Interpreting Performance

4.) Criterion-referenced

meaning is provided by describing
what students can and cannot do.

involve comparing a student's
performance to well-defined content
domain.



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Criterion-referenced

Examples of content domains include

- ✓ the ability to locate a word in a dictionary;
- ✓ to associate famous composers of music with historical periods;
- ✓ or to solve an algebraic equation involving whole numbers and one unknown.



Summary of the four frames of reference

Reference	Interpretation Provided by This Reference	Condition That Must be Present for This to be Useful
Ability-referenced	How are students performing relative to what they are capable of doing?	Requires good measures of what students are capable of doing; their maximum possible performance
Growth-referenced	How much have students changed or improved relative to what they were doing earlier?	Requires pre- and post measures of performance that are highly reliable
Norm-referenced	How well are students doing with respect to what is typical or reasonable?	To whom students are being compared must be clearly understood
Criterion-referenced	What can students do and not do?	Content domain that was assessed must be well defined



Determine which is ability, growth, norm and criterion referenced interpretation.

- Criterion **1.** John can shoot three baskets in a row.
- Growth **2.** Amy had a five point difference in a twenty item pre-test and post-test.
- Norm **3.** Carl was the top student of the class.
- Criterion **4.** Mark completed a research paper.
- Ability **5.** Harry finally passed the remedial test.



Criterion and Norm interpretations

- Both interpretations can be and usually are applied to a performance.
- It is possible for a test to be neither criterion nor norm-referenced.

Example. A student got a score of 25.
A score of 25 has no meaning when given without reference, such as what the test measured or how others did on the test.



Criterion and Norm interpretations

Classroom tests lack both criterion-referenced and norm-referenced interpretations.

- *Janet scored 92% on an algebra exam.*
- *Grace surpassed the criterion score 80% and passed the spelling test.*

The performance CANNOT be given a criterion-referenced interpretation unless SPECIFIC SKILLS measures are detailed.

Neither norm-referenced, unless COMPARISON scores are also given.



Does each of the following allow a criterion-referenced or norm-referenced interpretation of performance?

- Norm-referenced 1. Gina won the 100-meter race.
- Criterion-referenced 2. Arnold ran one mile in 5 minutes, 12 seconds.
- Norm-referenced 3. George scored near average on the final chemistry exam.
- Criterion-referenced 4. Carmen can separate alcohol from water through distillation.
- Criterion-referenced 5. In geometry, David cannot bisect an angle.
- Norm-referenced 6. Heidi obtained the best score on the geometry test.



Desired Qualities in Criterion and Norm-Referenced Tests

1.) Item difficulty

- difficulty of test items **affects the variability** of scores.
- It affects how adequately scores can be given a norm-referenced interpretation.
- When constructing norm-referenced tests, it is best to **avoid items** that will be **answered correctly** by all or none of the students.



Desired Qualities in Criterion and Norm-Referenced Tests

2.) Sensitivity to Instruction

- “the degree to which students' performances on a test accurately reflect the quality of instruction

- It is suggested that items used in *criterion-referenced tests* should be difficult to answer before instruction and easy to answer following instruction.



Desired Qualities in Criterion and Norm-Referenced Tests

3.) Number of Items

- criterion-referenced (focused domain) require **more items** than norm-referenced (broad content domain) testing.

- Use criterion-reference tests to measure a subset of critical skills

- Use norm-referenced tests to obtain comprehensive measures of student achievement.



Sources of Evaluative Information

1. Cumulative Record

Holds all information collected on students over the years such as vital statistics, academic records, conference information, health records, family data, scores on tests, behaviour and etc.



Sources of Evaluative Information

2. **Personal Contact**

Refers to the teacher's daily interactions with his/her students. Observational information through watching and listening to students in various situations.

Used to determine readiness to the next lesson or more guided activities.



Sources of Evaluative Information

3. **Analysis**

Teacher's analysis of error committed by students determines students' attitude and achievement

Used to identify students' learning difficulties.



Sources of Evaluative Information

4. Open-ended themes, Diaries and Journals

A technique used to make students write about their lives in and out of the school such as:

- a. What things do you like and dislike about school?
- b. What do you want to become when you grow up?

It provides valuable evaluative information for assessment.



Sources of Evaluative Information

5. **Conferences**

Parents provide valuable information which can explain why students are experiencing academic problems together with previous teachers and guidance counsellors.

Used to shed light on test results and personality factors, which might affect students' performance in class.



Sources of Evaluative Information

6. **Testing**

The most common measurement technique used in the classroom.

Used to measure student's cognitive achievement, as well their attitudes, values, feelings, and motor skills.



Principles of high quality assessment

- High quality classroom assessment is crucial to the improvement of teaching and learning that incorporates the four criteria of knowledge, skills, behavior and attitudes.





Principles of high quality assessments

1. **Clarity of Learning targets**

- Quality assessments begins with clear and appropriate learning targets which include both what students know and can do and the criteria for judging student performance.
 - Are the targets at the right difficulty to motivate students?
 - Is there adequate balance among different targets?
 - Are the targets comprehensive, covering all dimensions?



Principles of high quality assessments

2. Appropriateness of assessment methods

- It is best to match the method of assessment to learning targets.
- Know when to use the different types of classroom tests and assessments.



Principles of high quality assessments

3. **Validity**

- the degree to which score-based inference is appropriate, reasonable and useful.
- different types of evidence are used to establish the validity of classroom tests, the most important of which is content-related evidence.



Principles of high quality assessments

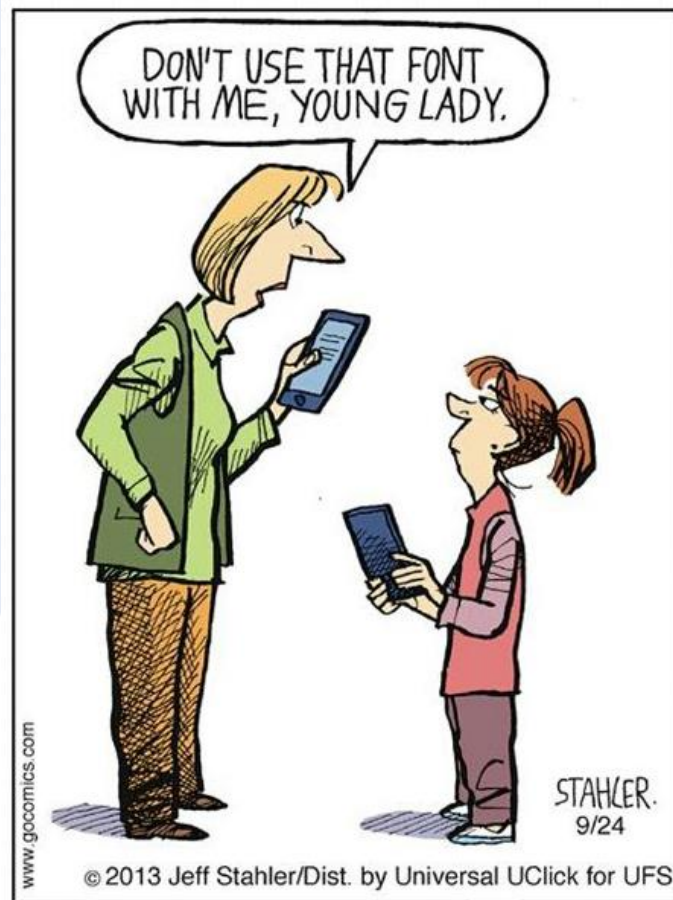
4. **Reliability**

- Used to measure the error in testing.
- Measures the degree of consistency when several items measure the same thing and stability when the same measures are given across time.
- Errors maybe internal “Fatigue, anxiety, health, mood” external “directions, item ambiguity, environment.”

Principles of high quality assessments

5. Fairness

- Assessment is fair if it is unbiased and provides students with a reasonable opportunity to demonstrate what they have learned.





Principles of high quality assessments

6. **Positive consequences**

- Positive consequences for both teachers and students enhance the overall quality of assessment, particularly the effect of the assessment on student motivation and study habits.
- Will the assessment help the teacher make more valid judgments, or will it tend to make judgments on students more difficult?



Principles of high quality assessments

7. Practicality and efficiency

- Assessments need to take into consideration the teacher's familiarity with the method, the time required, the complexity of administration, the ease of scoring and interpretation, and the cost to determine the assessments' practicality and efficiency.



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Instructional Objectives

- Describes an **observable event** that will indicate that a student has learned the targeted knowledge.
- A teacher CANNOT assess a student's learning unless there is no observable evidence of the learning.





Instructional Objectives

- Example:
 - How would students indicate that they know how the universe is large and contains large number of objects?
 - How would students indicate that they know the concept of multiplication?



- To assess a student's knowledge, a teacher must ask students to do something **observable/visible** that **indicates presence** of that knowledge.





- Instructional objectives represent **the teacher's starting point in instruction**:
 - What should students learn?
 - What skills and knowledge should be the focus of instruction?





Instructional objectives

- These are statements that **describe the abilities students should be able to display to demonstrate** that important concepts and principles have been incorporated into their knowledge.
- These statements **indicate what the learner should be able to do at the end of an instructional sequence.**
- These should be written in a form and at a level of specificity that will make them most useful for their intended purposes.



- For purpose of Planning
 - The objectives can be used to determine prerequisite knowledge learners need to enable them to benefit from instruction.





- For purpose of learner information
 - Students need to know what is expected of them.
 - To promote intentional learning rather than incidental.





- For purpose of **evaluation planning**
 - Objectives are used to **develop tools of performance assessment**.
 - It **dictates the type of measure** to use in assessing achievement





SMART Learning Objectives

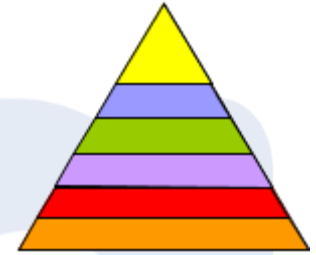
Specific	Is there a description of a precise behavior and the situation it will be performed in? Is it concrete, detailed, focused and defined?
Measurable	Can the performance of the objective be observed and measured?
Achievable	With a reasonable amount of effort and application can the objective be achieved? Are you attempting too much?
Relevant	Is the objective important or worthwhile to the learner or stakeholder? Is it possible to achieve this objective?
Time-bound	Is there a time limit, rate number, percentage or frequency clearly stated? When will this objective be accomplished?



Classification of Learning Outcomes

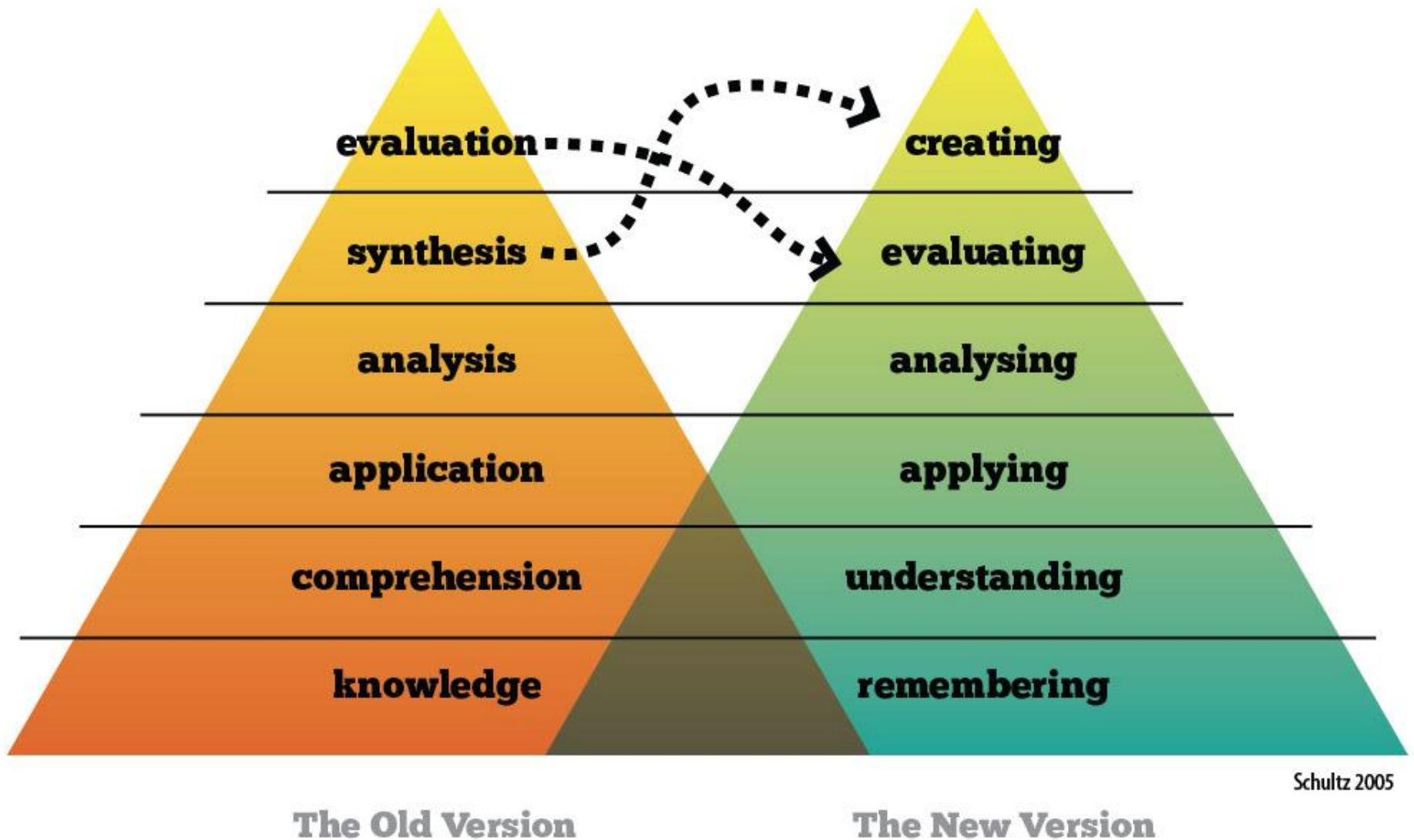
- Learning outcomes can be classified from general to particular, using certain division or domains of learning.
- Domains of learning designate a large group of learning opportunities, broad in scope, that are planned to achieve a single set of closely related educational goals.
- These domain classifications or taxonomies are developed by **Benjamin Bloom** and Associates.
- *Lorin. W. Anderson, a former student of Bloom and David. R. Krathwohl, one of the co-authors of Bloom's book, led a team of experts in revising Bloom's taxonomy.*

The Cognitive Domain

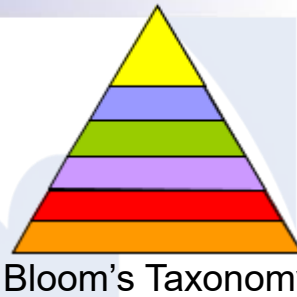


- Involves knowledge and the development of intellectual attitudes and skills
- A hierarchy of educational objectives
- Subdivisions ranging from the simplest behavior to the most complex

Bloom's Taxonomy – Anderson's Revision



Knowledge / Remembering



- **Knowledge/Remembering** is defined as the remembering of previously learned material.
- This may involve the recall of a wide range of material, from specific facts to complete theories, but all that is required is the bringing to mind of the appropriate information.
- **Learning outcomes:** Knowledge represents the lowest level of learning outcomes in the cognitive domain.
- Examples of learning objectives at this level are:
 - list common terms
 - define specific facts
 - enumerate methods and procedures
 - describe basic concepts
 - name principles



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Knowledge / Remembering



Bloom's Taxonom

- **Verb Cues:**

- List
- Define
- Tell
- Describe
- Identify
- Show
- Label
- Collect
- Examine
- Tabulate
- Quote
- Name

- Who
- When
- Where
- Match
- Select



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Understanding



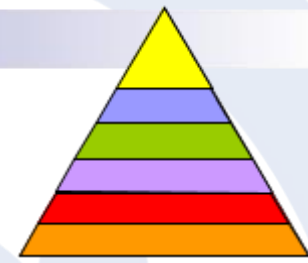
Bloom's Taxonomy

- **Understanding** is defined as the ability to grasp the meaning of material.
- This may be shown by translating material from one form to another (words to numbers), by interpreting material (explaining or summarizing), and by estimating future trends (predicting consequences or effects).
- These **learning outcomes** go one step beyond the simple remembering of material by illustrating understanding of meaning in concepts.



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Understanding



Bloom's Taxonomy

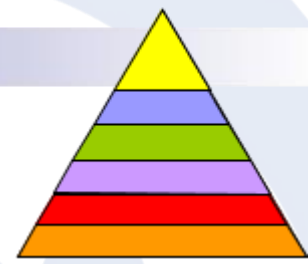
- **Verb Cues:**

- Summarize
- Describe
- Interpret
- Contrast
- Predict
- Associate
- Distinguish
- Estimate
- Differentiate
- Discuss
- Extend

- Examples of learning objectives at this level are:

- describe facts and principles
- interpret verbal material
- interpret charts and graphs
- translate verbal material to mathematical formulae
- estimate the future consequences implied in data
- justify methods and procedures

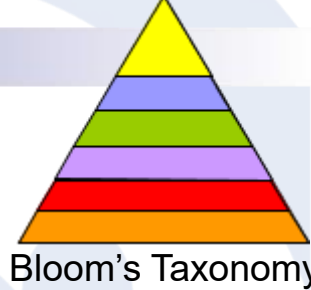
Applying



Bloom's Taxonomy

- **Applying** refers to the ability **to use learned material** in new and concrete situations.
- This may include the application of such things as rules, methods, concepts, principles, laws, and theories.
- **Learning outcomes** in this area require a higher level of understanding such as carrying out or using a procedure in a given situation.

Applying



- **Verb Cues:**

- Apply
- Demonstrate
- Calculate
- Complete
- Illustrate
- Show
- Solve
- Examine
- Modify
- Relate
- Change
- Classify
- Experiment
- Discover

- **Examples of learning objectives at this level are:**
 - apply concepts and principles to new situations
 - apply laws and theories to practical situations
 - solve mathematical problems
 - construct graphs and charts
 - demonstrate the correct usage of a method or procedure



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Analyzing



Bloom's Taxonomy

- **Analyzing** refers to the ability **to break down material** into its component parts so that its organizational structure may be understood. It is one of the key scientific instruments that we use in computer science.
- This may include the identification of parts, analysis of the relationship between parts, and recognition of the organizational principles involved.
- **Learning outcomes** here represent a higher intellectual level than comprehension and application because they require an understanding of both the content and the structural form of the material.



TEACHER

Analyzing



Bloom's Taxonomy

- **Verb Cues:**

- Analyze
- Separate
- Order
- Explain
- Connect
- Classify
- Arrange
- Divide
- Compare
- Select
- Explain
- Infer

- **Examples of learning objectives at this level are:**

- recognize unstated assumptions
- recognizes logical fallacies in reasoning
- distinguish between facts and inferences
- evaluate the relevancy of data
- analyze the organizational structure of a work (art, music, writing)



TEACHER

Evaluating



Bloom's Taxonomy

- **Evaluating** is concerned with the ability **to judge the value** of material (statement, novel, poem, research report) for a given purpose.
- The judgments are to be based on definite criteria. These may be internal criteria (organization) or external criteria (relevance to the purpose) and the student may determine the criteria or be given them.
- **Learning outcomes** in this area contain elements of all the other categories, plus conscious value judgments based on clearly defined criteria.



TEACHER

Evaluating



Bloom's Taxonomy

- **Verb cues**

- Assess
- Decide
- Rank
- Grade
- Test
- Measure
- Recommend
- Convince
- Select
- Judge
- Explain
- Discriminate

- Support
- Conclude
- Compare

- **Examples of learning objectives at this level are:**

- judge the logical consistency of written material,
- assess the adequacy with which conclusions are supported by data
- judge the value of a work (art, music, writing) by the use of internal criteria
- Measure the value of a work (art, music, writing) by use of external standards of excellence

Creating



- **Creating** refers to the ability to put parts together to form a new whole.
- This may involve the production of a unique communication (theme or speech), a plan of operations (research proposal), or a set of abstract relations (scheme for classifying information).
- **Learning outcomes** in this area are highest in the cognitive hierarchy because they stress creative behaviours, with major emphasis on the formulation of new patterns or structure.



TEACHER

Creating



Bloom's Taxonomy

- **Verb Cues:**

- Combine
- Integrate
- Modify
- Rearrange
- Substitute
- Plan
- Create
- Design
- Invent
- Compose
- Formulate
- Prepare
- Generalize
- Rewrite

- **Examples of learning objectives at this level are:**

- write a well organized theme
- compose a well organized speech
- writes a creative short story (or poem or music)
- propose a plan for an experiment
- integrate learning from different areas into a plan for solving a problem
- formulates a new scheme for classifying objects (or events, or ideas)



Think Critically

Determine if the following is A. Remembering, B. Understanding, C. Applying, D. Analyzing, E. Evaluating, F. Creating.

- C** 1. Solve for the value in the given equation.
- B** 2. Summarize the story using your own words
- E** 3. Critique the worth of an artwork
- A** 4. State the colors of the rainbow
- D** 5. Explain the scientific process of investigation

Write your answer on the meeting chat.

Affective Domain

- Emanates from the emotional component of behavior where the concern is on **how the learner feels**.
- Covers sensing, feeling or believing.

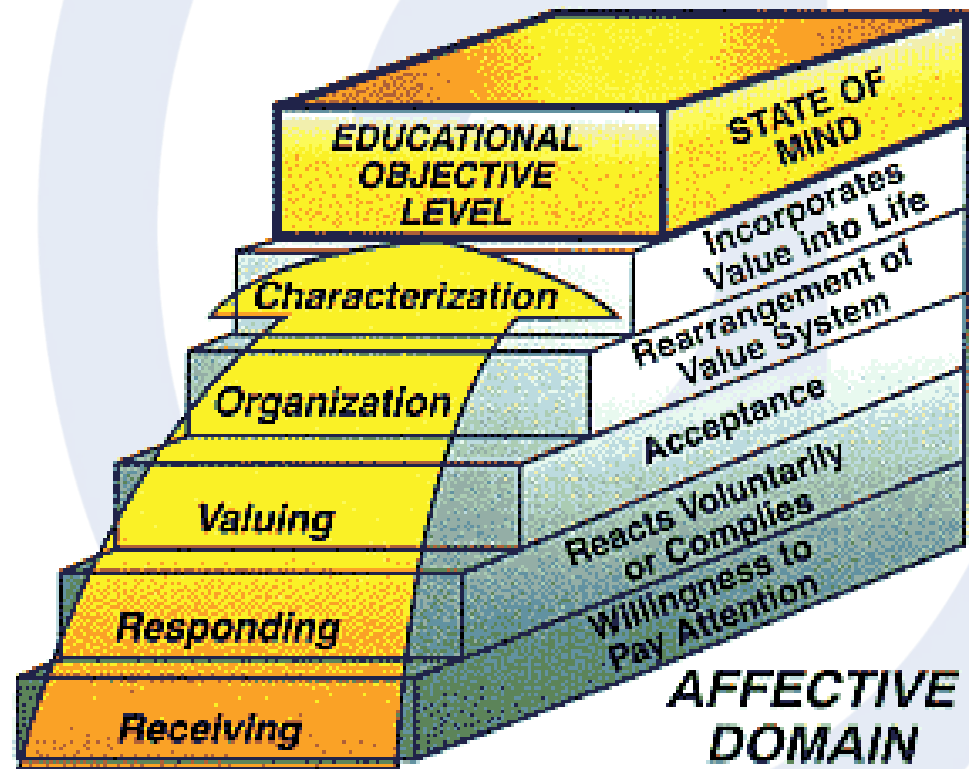


Figure 1-5. D. R. Krathwohl's hierarchical taxonomy for the affective domain (attitudes, beliefs, and values) contains five educational objective levels.



RECEIVING

- Refers to becoming **aware** of an idea, process or thing and is willing to notice a particular phenomenon.

- ask, attend, be aware, choose, describe, follow, give, hold, identify, listen, locate, name, receive, reply, select, show alertness, tolerate, use, view, watch

Ex. During an examination, the students will be able to **follow instructions** properly in answering the test.



RESPONDING

- Refers to **active participation** on the part of the student. The student is sufficiently motivated not to just be willing to attend, but is **actively attending**.



agree (to), answer, ask, assist, communicate, comply, consent, conform, contribute, cooperate, discuss, follow-up, greet, help, indicate, inquire, label, obey, participate, pursue, question, react, read, reply, report, request, respond, seek, select, visit, volunteer, write

Ex. During a graded group work, the students will be able to **participate** by giving his opinion on the assigned topic.



TEACHER

VALUING

- The student attaches, applies or show a **value** to an object, phenomenon, or piece of information.
- accept, adopt, approve, complete, choose, commit, describe, desire, differentiate, display, endorse, exhibit, explain, express, form, initiate, invite, join, justify, prefer, propose, read, report, sanction, select, share, study, work

Topic : Democracy

Given a quiz, the students will be able to **demonstrate belief** in the democratic process by giving at least three advantages of a democratic form of government.



ORGANIZATION

Bringing together a **complex** of **values**, possible disparate values, resolving conflicts between them, and beginning to build an internally consistent value system.

■ adapt, adhere, alter, arrange, categorize, classify, combine, compare, complete, defend, explain, establish, formulate, generalize, group, identify, integrate, modify, order, organize, prepare, rank, rate, relate, synthesize, systemize

Topic : Democracy

By **writing an article** about the student council , the students will be able to **recognize the need for balance** between freedom and responsibility .



TEACHER

CHARACTERIZATION

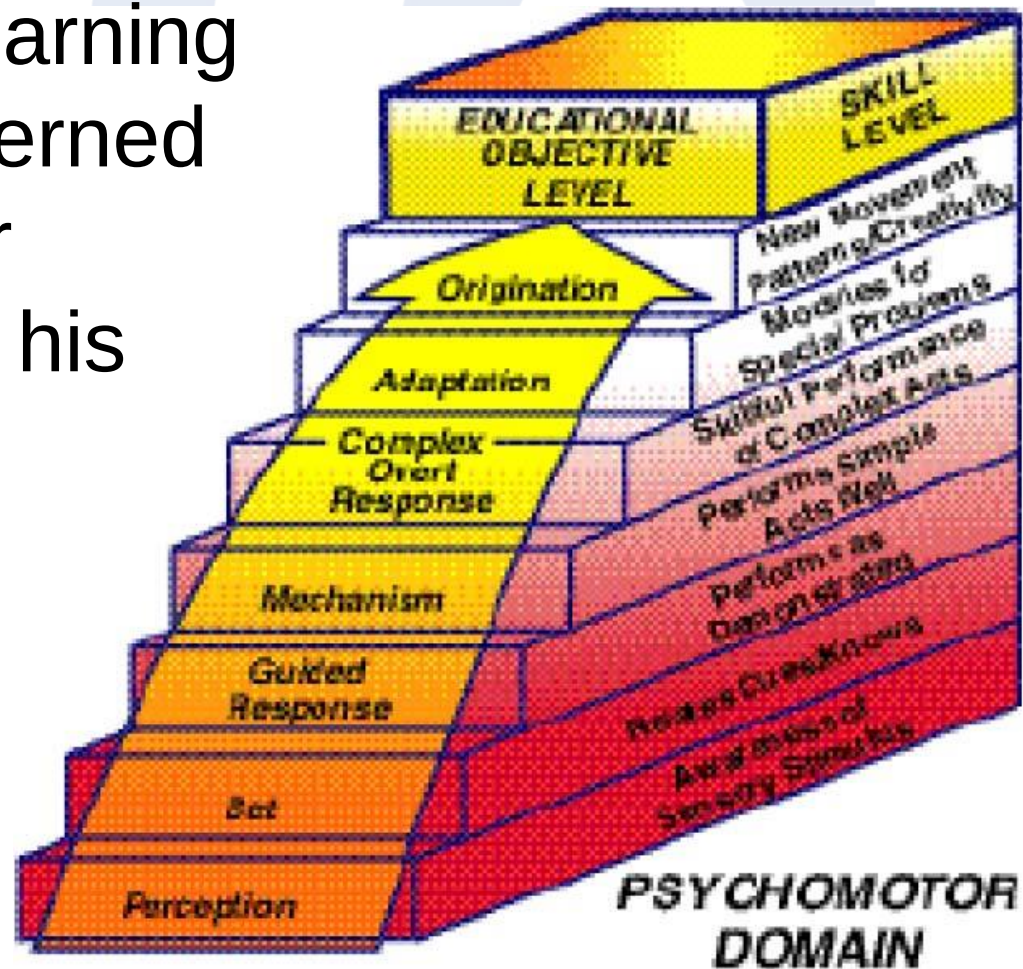
- The student has held a particular value or belief that now exerts **influence on their behavior** so that it becomes a characteristic.

Ex. Write an essay on how democracy influence one's philosophy.

- act, advocate, behave, characterize, conform, continue, defend, devote, disclose, discriminate, display, encourage, endure, exemplify, function, incorporate, influence, justify

Psychomotor Domain

- The category of learning objectives is concerned with how a learner controls or moves his body.





TEACHER

PERCEPTION

- The ability to **use sensory cues** to guide motor activity.
- This ranges from sensory stimulation, through cue selection, to translation.



TEACHER

PERCEPTION

● Examples:

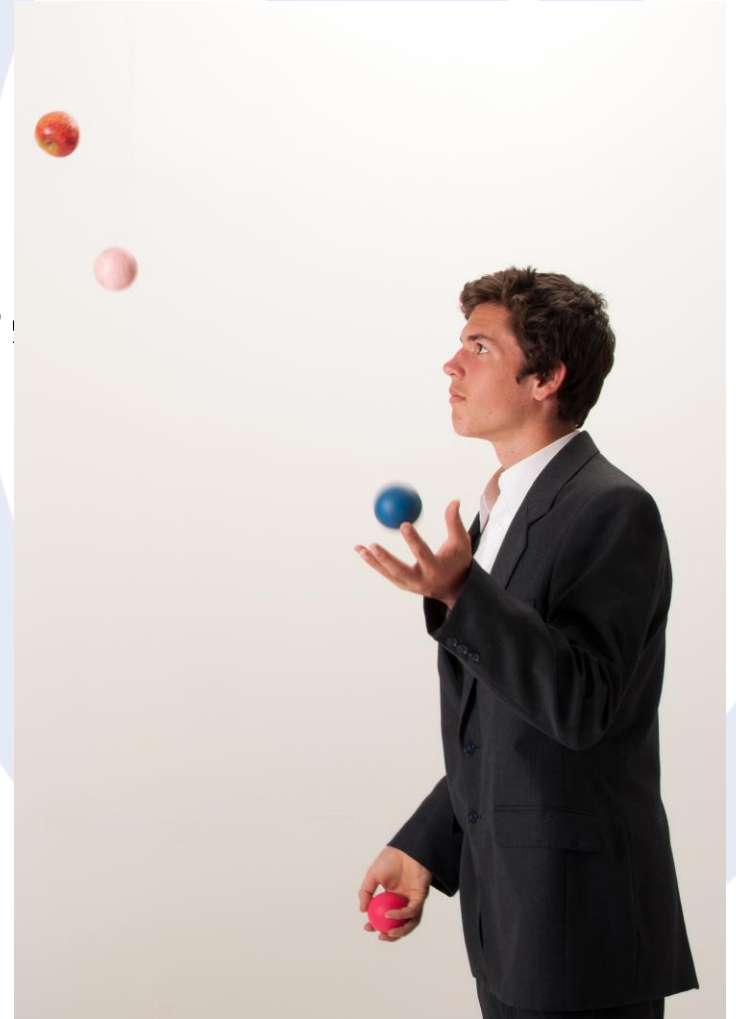
- Detects non-verbal communication cues.
- Estimate where a ball will land after it is thrown and then moving to the correct location to catch the ball.
- Adjusts heat of stove to correct temperature by smell and taste of food.
- Adjusts the height of the forks on a forklift by comparing where the forks are in relation to the pallet.
- “By the end of the music theatre program, students will be able to relate types of music to particular dance steps.”



TEACHER

PERCEPTION

- chooses, describes, detects, differentiates, distinguishes, feels, hears, identifies, isolates, notices, recognizes, relates, selects, separates, touches





SET

- **Readiness to act.**
- It includes mental, physical, and emotional sets.
- These three sets are dispositions that predetermine a person's response to different situations (sometimes called mindsets).



SET

- Examples:
 - Knows and acts upon a sequence of steps in a manufacturing process.
 - Recognize one's abilities and limitations.
 - Shows desire to learn a new process (motivation).
 - “By the end of the physical education program, students will be able to demonstrate the proper stance for batting a ball.”

SET

- arranges, begins, displays, explains, gets set, moves, prepares, proceeds, reacts, shows, states, volunteers, responds, starts





TEACHER

GUIDED RESPONSE

- The **early stages in learning a complex skill** that includes imitation and trial and error.
- Adequacy of performance is achieved by practicing.



GUIDED RESPONSE

- Examples:
 - Performs a mathematical equation as demonstrated.
 - Follows instructions to build a model.
 - Responds hand-signals of instructor while learning to operate a forklift.
 - “By the end of the physical education program, students will be able to perform a golf swing as demonstrated by the instructor.”



TEACHER

GUIDED RESPONSE

- assembles, builds, calibrates, constructs, copies, dismantles, displays, dissects, fastens, fixes, follows, grinds, heats, imitates, manipulates, measures, mends, mixes, reacts, reproduces, responds sketches, traces, tries.





TEACHER

MECHANISM

- This is the **intermediate stage in learning a complex skill.**
- Learned responses have become habitual and the movements can be performed with some confidence and proficiency.



MECHANISM

- Examples:
 - Use a personal computer.
 - Repair a leaking faucet.
 - Drive a car.
 - “By the end of the biology program, students will be able to assemble laboratory equipment appropriate for experiments.”



TEACHER

MECHANISM

- assembles, builds, calibrates, completes, constructs, dismantles, displays, fastens, fixes, grinds, heats, makes, manipulates, measures, mends, mixes, organizes, performs, shapes, sketches.





COMPLEX OVERT RESPONSE

- The **skillful performance of motor acts** that involve complex movement patterns.
- Proficiency is indicated by a quick, accurate, and highly coordinated performance, requiring a minimum of energy.



COMPLEX OVERT RESPONSE

- Examples:
 - Maneuvers a car into a tight parallel parking spot.
 - Operates a computer quickly and accurately.
 - Displays competence while playing the piano.
 - “By the end of the industrial education program, students will be able to demonstrate proper use of woodworking tools to high school students.”



COMPLEX OVERT RESPONSE

- assembles, builds, calibrates constructs, coordinates, demonstrates, dismantles, displays, dissects, fastens, fixes, grinds, heats, manipulates, measures, mends, mixes, organizes, sketches.





TEACHER

ADAPTATION

- Skills are well developed and the individual can **modify movement patterns** to fit special requirements.
- alter response to reliably meet varying challenges



ADAPTATION

- Examples:
 - Responds effectively to unexpected experiences.
 - Modifies instruction to meet the needs of the learners.
 - Perform a task with a machine that it was not originally intended to do.
 - “By the end of the industrial education program, students will be able to adapt their lessons on woodworking skills for disabled students.”



ADAPTATION

- adapts, adjusts, alters, changes, integrates, rearranges, reorganizes, revises, solves, varies.





ORINATION

- **Creating new movement patterns** to fit a particular situation or specific problem.
- Learning outcomes emphasize creativity based upon highly developed skills.



TEACHER

ORIGINATION

- develop and execute new integrated responses and activities
- Examples:
 - Constructs own style of playing.
 - Develops a new and comprehensive training programming.
 - Creates a new gymnastic routine.



ORINATION

- arranges, builds, combines, composes, constructs, creates, designs, formulates, initiate, makes, modifies, originates, re-designs, trouble-shoots.



Origination: Creating new movement patterns to fit a particular situation

Adaptation: Skills well developed & can be modified to fit special requirements

Complex Overt Response: The skillful performance of complex movements

Mechanism: Learned responses have become habitual

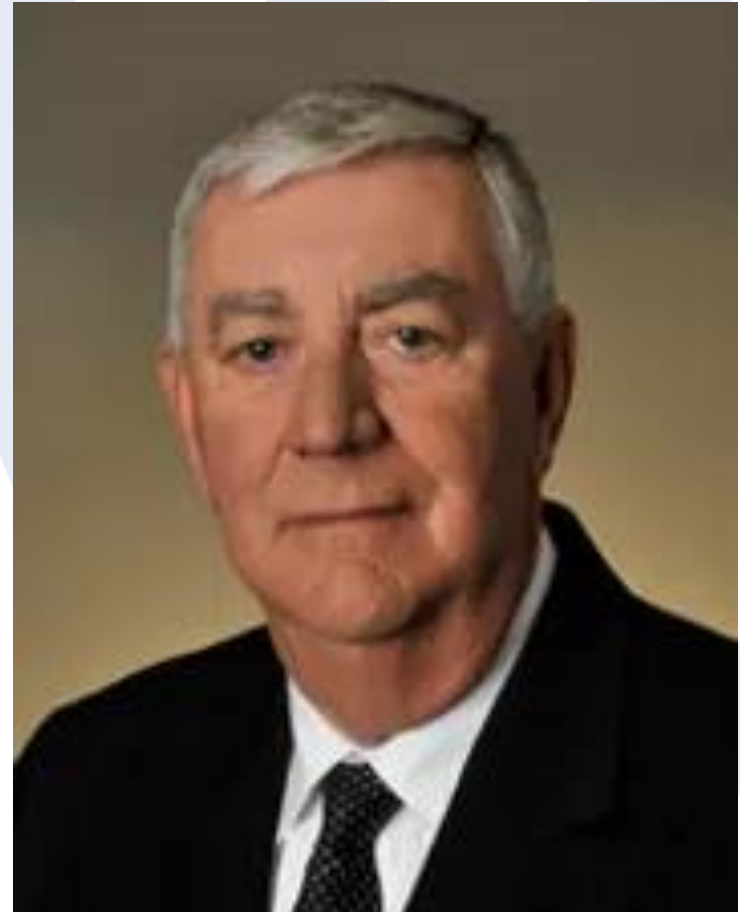
Response: Early stages in learning a complex skill - imitation and trial and error

Set: Readiness to act

Perception: Ability to use sensory cues to guide motor activity

Gagne's Categories of Learning Outcomes

- Robert Gagne (1985) educational psychologist classified learning outcomes into five categories
- The Gagné taxonomy is perhaps the most popular of the many learning taxonomies in the field of instructional design





Categories of Learning Outcomes

- Student performance can be categorized into capabilities.
 - 1.) **Declarative Knowledge**
 - refers to the information
 - 2.) **Procedural knowledge** has four categories:
 - Discriminations, concepts, rules and complex skills.
 - 3.) **Motor skills** – bodily movements
 - 4.) **Attitudes** – internal state which affect choice of action



TEACHER

1.) Declarative Knowledge

- It refers to information that can be stated verbally.
- Includes the recall of specific facts, principles, trends, criteria and ways of organizing events.
- Asking students to state what they know measures declarative knowledge.

Example

1. What is the difference between air and oxygen?
2. Why do interest rates affect the stock market?
3. Contrast the geography of north and south Africa.
4. Why do heavy objects like ships float?



2.) Procedural knowledge

- Knowledge of how to do things.
- Includes demonstrating conversions between the Fahrenheit and Celsius scales; classifying whales; identifying objects in a picture.
- Referred to as intellectual skills.

Categories	Bloom's Taxonomy	Gagne's Capabilities
Declarative knowledge refers to information that can be stated verbally	Knowledge/ Remembering: Information such as specific facts, principles, trends, criteria, and ways of organizing events.	Verbal information:
Procedural knowledge Knowledge of how to do things	Intellectual skills Understanding/ Comprehension Applying/ Application Analyzing/ Analysis Evaluating/ Evaluating Creating/ Synthesis	Intellectual skills Discriminations Concrete Concepts Defined concepts Rules Higher Order Rules



Determine if the following indicates a **declarative** or **procedural** knowledge.

Ex. Knowing that a touchdown in football is worth 6 points. **declarative**

- 1.) Distinguishing between a touchdown and a safety. **procedural**
- 2.) Looking at the fish caught by each of several individuals and determining who caught the most fish. **procedural**
- 3.) Recalling the next day who caught the most fish. **declarative**
- 4.) Naming the capital cities of each state. **declarative**
- 5.) Given descriptions of several cities, identifying which are capital cities. **procedural**

Refresher Course

Assessment in Learning

- Part 1 - November 2022
- *Dr. Stephen Jay D. Co*